

SciencesPo

**INTERMEDIATE
MACROECONOMICS
LESSON 4
BALANCED GROWTH PATH**

FALL 2025, LE HAVRE CAMPUS

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Logarithm Rule

- If $z = xy$ then:

$$g_Z = g_X + g_Y$$

- Proof: Let Δz be the increase in z when x increases by Δx and y increases by Δy . Then, by definition:

$$z + \Delta z = (x + \Delta x)(y + \Delta y)$$

- Divide both sides by z .

$$\begin{aligned}\frac{z + \Delta z}{z} &= \left(1 + \frac{\Delta z}{z}\right) = \\ &= \frac{(x + \Delta x)(y + \Delta y)}{xy} = \left(1 + \frac{\Delta x}{x}\right) \left(1 + \frac{\Delta y}{y}\right) \\ &= (1 + g_X)(1 + g_Y)\end{aligned}$$

$$\frac{(1+x)}{(1+y)} \approx 1+x-y$$

Proof: consider the product of $(1+x-y)(1+y) = 1+x+xy-y^2$

If both x and y are small number then, $xy - y^2$ would be small too

Thus, $(1+x-y)(1+y) = 1+x$

Divide both side of this approximation by $(1+y)$ gives the

$$\frac{(1+x)}{(1+y)} \approx 1+x-y$$

$$(1 + g_X)(1 + g_Y) = 1 + g_X + g_Y$$

$$\text{Thus } 1 + \frac{\Delta z}{z} = (1 + g_Z)$$

$$(1 + g_Z) = 1 + g_X + g_Y$$

$$(g_Z) = g_X + g_Y$$

Characteristics of Balance of Growth Rate at Steady State

Variable		Growth Rate
Capital per effective worker	$\hat{k}$	0
Output per effective worker	$\hat{y}$	0
labor	N	g_N
capital	K	$g_A + g_N$
output	Y	$g_A + g_N$
Capital per worker	k	g_A
Output per worker	y	g_A

On the balanced growth path (Steady state/long run)

$$\Delta \hat{k} = sf(\hat{k}) - (\delta + g_A + g_N)\hat{k} = 0$$

- $g(Y)=?$

- At steady state $g(\hat{y}) = g\left(\frac{Y}{AN}\right) = 0$

- $g(\hat{y}) = g\left(\frac{Y}{AN}\right) = g(Y) - g(AN) = 0$

$$\mathbf{g}(Y) = \mathbf{g}(A) + \mathbf{g}(N)$$

- $g(K)=?$

- At steady state $g(\hat{k}) = g\left(\frac{K}{AN}\right) = 0$

- $g(\hat{k}) = g\left(\frac{K}{AN}\right) = g(K) - g(AN) = 0$

$$g(K) - g(AN) = g(K) - [g(A) + g(N)] = 0$$

$$\mathbf{g}(K) = [\mathbf{g}(A) + \mathbf{g}(N)]$$

- $g(y)=?$
- $g(y) = g\left(\frac{Y}{N}\right) = g(Y) - g(N) = \{g(A) + g(N)\} - g(N) = g(A)$
- $g(k)=?$

Variable		Growth Rate
Capital per effective worker	$\hat{k}$	0
Output per effective worker	$\hat{y}$	0
labor	N	g_N
capital	K	$g_A + g_N$
output	Y	$g_A + g_N$
Capital per worker	k	g_A
Output per worker	y	g_A

Example Problem for Problem Set 2

Suppose that an economy's production function is

$$Y = \sqrt{K}\sqrt{AN}$$

Suppose that the saving rate, s , is equal to 17%, and that the rate of depreciation (δ) is equal to 9%. Suppose further that the number of workers grows at 2% per year and that the rate of technological progress is 2% per year. Using this set-up, compute the following:

- (a) The steady-state value of the capital stock per effective worker
- (b) The steady-state value of the output per effective worker
- (c) The steady-state value of the growth rate of output per effective worker
- (d) The steady-state value of the growth rate of output per worker
- (e) The steady-state value of the growth rate of output
- (f) Suppose that the rate of technological progress doubles to 4% per year. Recompute the values for (a)-(e)

- Q(a):

$$Y = \sqrt{K}\sqrt{AN}$$

$$\frac{Y}{AN} = \sqrt{\frac{K}{AN}} \sqrt{\frac{AN}{AN}} = \sqrt{\frac{K}{AN}}$$

$$\bullet \hat{y} = \sqrt{\hat{k}}$$

$$\text{At steady-state, } \Delta \hat{k} = sf(\hat{k}) - (\delta + g_A + g_N)\hat{k} = 0$$

$$sf(\hat{k}) = (\delta + g_A + g_N)\hat{k}$$

$$0.17 * \sqrt{\hat{k}} = (0.09 + 0.02 + 0.02)\hat{k}$$

$$0.17 * \sqrt{\hat{k}} = (0.13)\hat{k}$$

$$\frac{0.17}{0.13} = \frac{\hat{k}}{\sqrt{\hat{k}}}$$

$$\frac{0.17}{0.13} = \sqrt{\hat{k}}$$

$$(1.307)^2 = \left(\frac{\hat{k}}{\sqrt{\hat{k}}}\right)^2$$

$$\hat{k} = 1.71$$

$$\hat{k}^* = 1.71$$

- Q (b)

$$\hat{y}^* = \sqrt{\hat{k}^*} = 1.307$$

- Q(c):
- On the balanced growth path (steady state / long run): Capital per effective worker and output per worker are constant

$$\Delta \hat{k} = 0$$

$$\Delta \hat{y} = 0$$

So the answer is 0%

- Q1(d): 4pt
- At the steady-state, the growth rate of output per worker $g(y)$

$$g(y) = g\left(\frac{Y}{N}\right) = g(Y) - g(N) = \{g(A) + g(N)\} - g(N) = g(A) = 2\%$$

- Q(e):

$$g(Y) = g(A) + g(N) = 2\% + 2\% = 4\%$$

- $\Delta \hat{y} = 0$; $g(\hat{y}) = 0$ from Q1(c):

$$g(\hat{y}) = g\left(\frac{Y}{AN}\right) = g(Y) - g(AN) = g(Y) - \{g(A) + g(N)\} = 0$$
$$g(Y) = \{g(A) + g(N)\}$$

- Q(f)
- Change the $g(A)$ from 2% to 4% and do it again from (a) to (e)

$$Y = \sqrt{K} \sqrt{AN}$$

$$\frac{Y}{AN} = \sqrt{\frac{K}{AN}} \sqrt{\frac{AN}{AN}} = \sqrt{\frac{K}{AN}}$$

$$\hat{y} = \sqrt{\hat{k}}$$

At steady-state, $\Delta \hat{k} = sf(\hat{k}) - (\delta + g_A + g_N)\hat{k} = 0$

$$sf(\hat{k}) = (\delta + g_A + g_N)\hat{k}$$

$$0.17 * \sqrt{\hat{k}} = (0.09 + 0.04 + 0.02)\hat{k}$$

$$0.17 * \sqrt{\hat{k}} = (0.15)\hat{k}$$

$$\frac{0.17}{0.15} = \frac{\hat{k}}{\sqrt{\hat{k}}}$$

$$(1.3333)^2 = \left(\frac{\hat{k}}{\sqrt{\hat{k}}} \right)^2$$

$$\hat{k} = 1.28$$

$$\hat{k}^* = 1.28$$

$$\hat{y}^* = \sqrt{\hat{k}^*} = 1.13$$

- On the balanced growth path (steady state / long run): Capital per effective worker and output per worker are constant

$$\Delta \hat{k} = 0$$

$$\Delta \hat{y} = 0$$

So the answer is 0%

At the steady-state, the growth rate of output per worker $g(y)$

$$g(y) = g\left(\frac{Y}{N}\right) = g(Y) - g(N) = \{g(A) + g(N)\} - g(N) = g(A) = 4\%$$

$$g(Y) = g(A) + g(N) = 2\% + 4\% = 6\%$$

How to plot a graph on Excel

- Let's work on how to show the Solow residual as a data exercise!
- Please go to the data in the section of the Moodle Lesson 4